LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
Observation Task No.	2
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1. TITLE

Design and Implementation of a Technology Learning Page Using HTML5, CSS3 and Canvas

2. AIM

To design a styled webpage using HTML5 and CSS3 and to demonstrate HTML5 Canvas drawing operations such as a rectangle, circle, line, and text using JavaScript.

3. OBJECTIVE

1. To create a structured webpage using HTML5 elements.
2. To apply CSS styling using gradients, shadows, borders, spacing, and hover effects.
3. To create navigation links and content cards.
4. To use the HTML5 canvas element for drawing graphics.
5. To draw a rectangle, circle, line, and text using JavaScript.
6. To understand basic interaction between HTML, CSS, and JavaScript.

4. THEORY

HTML5 provides the basic structure of a webpage, while CSS3 is used to control its appearance. The Canvas API provides a drawing surface that can be controlled using JavaScript. The **getContext('2d')** method obtains a two-dimensional drawing context. Methods such as **fillRect()**, **arc()**, **moveTo()**, **lineTo()**, and **fillText()** are used to draw shapes and text. CSS pseudo-class **:hover** can be used to change the appearance of navigation links and buttons when the pointer moves over them.

Application Flow: HTML Structure → CSS Styling → Canvas Element → JavaScript Canvas API → Shapes & Text → Styled Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML document and set the title as 'Technology Learning Page'.
2. Link the external stylesheet style.css.
3. Create a header containing the webpage title.
4. Create a navigation bar with Dashboard, Courses, and Projects links.
5. Create a canvas element with an ID named drawingArea.
6. Create three content boxes for Python, Database, and AI.
7. Add descriptions and Explore buttons to the content boxes.
8. Create a footer with copyright information.
9. Use JavaScript to access the canvas and obtain the 2D drawing context.
10. Draw a blue rectangle, a circle, a line, and the text 'JATIN' on the canvas.

11. Open the webpage in a browser and verify the styling and canvas output.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>

<head>
    <title>Technology Learning Page</title>
    <link rel="stylesheet" href="style.css">
</head>

<body>

<header>
    <h1>Technology Learning Page</h1>
</header>

<nav>
    <a href="#">Dashboard</a>
    <a href="#">Courses</a>
    <a href="#">Projects</a>
</nav>

<canvas id="drawingArea" width="500" height="220"></canvas>

<section class="content">

    <div class="box">
        <h2>Python</h2>
        <p>Python is used for programming and application development.</p>
        <button>Explore</button>
    </div>

    <div class="box">
        <h2>Database</h2>
        <p>Databases are used to store and manage information.</p>
        <button>Explore</button>
    </div>

    <div class="box">
        <h2>AI</h2>
        <p>Artificial Intelligence helps machines perform intelligent tasks.</p>
        <button>Explore</button>
    </div>

</section>

<footer>
    <p>© 2026 Technology Learning Page</p>
</footer>

<script>

let canvas = document.getElementById("drawingArea");
let ctx = canvas.getContext("2d");

ctx.fillStyle = "blue";
ctx.fillRect(20, 20, 120, 70);

ctx.beginPath();
ctx.arc(220, 55, 35, 0, 2 * Math.PI);
ctx.stroke();

ctx.beginPath();
ctx.moveTo(20, 130);
ctx.lineTo(300, 130);
ctx.stroke();

ctx.font = "28px Arial";
ctx.fillText("JATIN", 170, 190);

</script>

</body>
</html>
```

style.css

```
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: #eef2f3;
}

header {
    background: linear-gradient(to right, #ff6600, #ff0066);
    color: white;
    text-align: center;
    padding: 25px;
    text-shadow: 2px 2px 5px gray;
}
```

```css
nav {
    background: #222;
    text-align: center;
    padding: 15px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 20px;
    font-size: 18px;
}

nav a:hover {
    color: orange;
}

canvas {
    display: block;
    margin: 25px auto;
    background: white;
    border: 2px solid #555;
}

.content {
    display: flex;
    justify-content: center;
    gap: 20px;
    margin: 30px;
}

.box {
    background: white;
    width: 250px;
    padding: 20px;
    border: 2px solid #777;
    border-radius: 12px;
    text-align: center;
    box-shadow: 5px 5px 10px #999;
}

button {
    background: #0066cc;
    color: white;
    border: none;
    padding: 10px 15px;
    border-radius: 5px;
    cursor: pointer;
}

button:hover {
    background: #ff6600;
}

footer {
    background: #222;
    color: white;
    text-align: center;
    padding: 15px;
    margin-top: 20px;
}
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<canvas>	Creates a drawing area that can be controlled using JavaScript.
getContext('2d')	Obtains the two-dimensional drawing context of the canvas.
fillRect()	Draws a filled rectangle on the canvas.
arc()	Creates a circular or arc path.
moveTo() and lineTo()	Defines the starting and ending points of a line.
fillText()	Draws text on the canvas.
linear-gradient()	Creates a smooth color transition for the header.
:hover	Changes the appearance of links or buttons when the pointer is over them.
display: flex	Arranges the content boxes horizontally using Flexbox.
box-shadow	Adds a shadow around the content boxes.

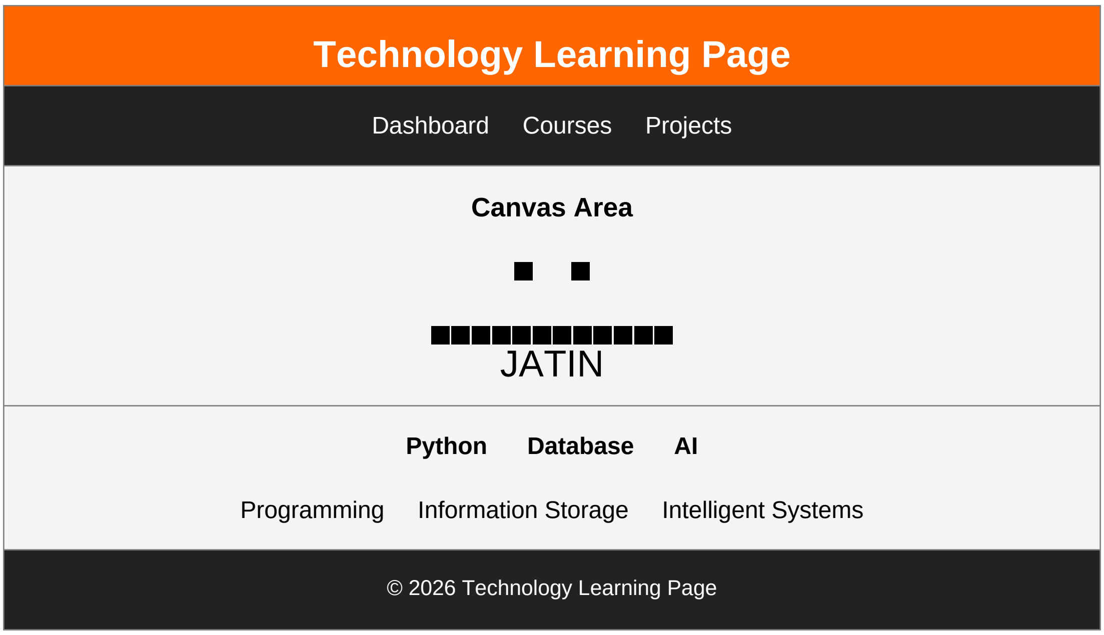
8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open webpage	Header and navigation display correctly	Displayed correctly	Pass
TC-02	Inspect canvas	Canvas area appears below navigation	Canvas displayed	Pass
TC-03	Inspect canvas drawings	Rectangle, circle, line and text appear	Drawings displayed	Pass
TC-04	Inspect content boxes	Three technology boxes are displayed	Boxes displayed correctly	Pass
TC-05	Move pointer over links/buttons	Hover style changes the element	Hover effects work	Pass

9. OUTPUT

Output 1: Technology Learning Page displaying a styled header, navigation bar, canvas, three technology cards, and footer.

Canvas Output: A blue rectangle, outlined circle, horizontal line, and the text 'JATIN' are drawn using the JavaScript Canvas API.



10. RESULT

Thus, the Technology Learning Page was successfully designed and implemented using HTML5, CSS3, and JavaScript. The webpage was verified for CSS styling, navigation, Flexbox layout, hover effects, and Canvas API drawing operations including rectangle, circle, line, and text.